KARAN SINGH GURJAR

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SUMMARY

Computer Science Engineer specializing in **AI, Diffusion Models, Remote Sensing, and Full-Stack Development**. Experienced in building high-performance deep learning pipelines, multimodal generative models, and production-ready applications. Strong problem-solver with a passion for research-driven ML systems.

EDUCATION

Master of Technology, Reva University — Expected 2026
Cgpa:7.12

Bachelor of Computer Science, Shri Ram group of colleges, Cgpa: 7.92 — 2020 - 2024

SKILLS

Programming: Python, Java
ML/DL: PyTorch, TensorFlow, Librosa, Pandas, Folium, iCrawler
Frameworks: ReactJS, FastAPI, Flask, Bootstrap
Databases: MySQL, MongoDB
DevOps: Linux, Docker, Kubernetes
Tools: Git, GitHub
Cloud/Compute: CUDA, Remote GPU workflows

EXPERIENCE

Full-Stack Web developer, Founding Engineer IT Dept LINK — Aug 2025 - Jan 2026
Indus Pharma India Pvt Ltd — *Neemrana, RJ*
One of the first two founding engineers leading digital transformation.
Designed and developed the company's first official web platform from scratch.
Improved outreach by **5000+ potential clients within the first quarter**.

PROJECTS

Bird Sound classification with geospatial visualization
Developed an ML-based system that classifies bird species from audio recordings and displays their geographical locations.
Trained a ResNet-18 model on a dataset of over 22,000 bird sound clips, achieving 53% classification accuracy.
Research paper published at the 2025 IEEE International Conference on Emerging Trends in Engineering and Computing (ETECOM) – Artificial Intelligence, Machine Learning, and Business Analytics Track.

Task-Aware SAR-to-RGB Image Translation (Diffusion + Transformers)
Remote Sensing • Deep Learning • Generative Modeling

- Built a multimodal framework converting SAR imagery into realistic RGB optical images.
- Designed a task-aware architecture combining **Attention U-Net, Swin Transformer, land-cover semantic fusion, and wavelet-based frequency enhancement**.
- Implemented **conditional diffusion** with *self-conditioning*, *classifier-free guidance*, and *dynamic thresholding* for stable iterative generation.
- Evaluated using **PSNR, SSIM, LPIPS**, and downstream land-cover accuracy on **SEN12MS**.
- Outperformed CNN and GAN baselines with higher semantic retention.

Hugging Face Hackathon (2025)

- Participated in the global Hugging Face community hackathon.